Predicting Order-Disorder Phase Transitions with Machine Learning in the Vicsek Flocking Model

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***Abstract*—In this study, we use machine learning to classify and interpolate the phase structure of the Vicsek flocking model across the three-dimensional parameter space (η, ρ, v_0). We construct a dataset of simulated parameter points and characterize each point using long-time dynamical observables. These observables are then used as inputs to a K-Means clustering procedure, which assigns each point to a disorder, order, or coexistence phase. Using these clustered labels, we train a neural-network classifier to learn the mapping from model parameters to phase behavior, achieving a classification accuracy of 0.92. The resulting phase map resolves a narrow coexistence region separating the ordered and disordered phases and extends the inferred phase boundaries beyond the originally sampled simulation points. More broadly, this approach provides a systematic way to convert sparse simulation data into a global phase diagram for collective-motion models.**

I. Introduction

Collective motion, or the spontaneous movement of many self-propelled particles [1], [2], can be observed throughout our universe in many different contexts, from bird flocking [3] to movement of bacterial colonies [4] to pedestrian traffic [5]. Across these situations, we can detect similar properties in the movement of these particles, otherwise known as active matter, such as spontaneous organization and emergent patterns across the population. The ubiquity of collective motion and the similarities between different instances of collective motion in diverse environments thus imply a possible set of consistent underlying universal rules [6]. As such, understanding these mechanisms behind collective motion can allow us to comprehend our world more fully, better analyze synchronized motion in nature, and improve modern swarming technologies.

Agent-based models (ABMs) are powerful tools for modeling complex systems and environments over time [7], [8]. ABMs are mechanistic simulations composed of individual, autonomous agents that are assigned certain attributes and behaviors. Each agent, which can represent a particle, human being, animal, or other individual unit, may interact with other agents or the environment, which produces emergent phenomena across the population. ABMs are particularly useful for observing behaviors only producible from a large population, as we can observe patterns across the population not explicitly programmed into the individual agents.

One such agent-based model of particular significance is the Vicsek model [9], proposed in 1995 by Tamás Vicsek, a foundational mathematical agent-based model used in the study of collective motion and active matter. Proposed as a model for the familiar behavior of bird flocking, the Vicsek model depicts the inception of collective motion from complete disorder in a far-from-equilibrium system, where momentum from particle interactions is not conserved [10]. However, the true distinctiveness of this model comes from its minimality [2]. As the Vicsek model contains only one rule, it is seen as one of the simplest agent-based models for collective motion [11]. Despite this, the simplicity of the Vicsek model masks a complex variety of emergent behaviors [12]–[14].

One of the most interesting and intricate behaviors of the Vicsek model is the non-equilibrium transition to collective motion from initial disorder. The subject of the phase transitions is highly controversial; the nature of the transition to ordered motion was originally thought to be a continuous second-order transition [9], which was then challenged a decade later and argued to be a first-order transition [2]. Later, the existence of a coexistence phase of traveling order bands was discovered through the reframing of the transition as a liquid-gas transition [12], with the coexistence phase then refactored into an additional cross-sea phase exhibiting an independent pattern for large system sizes [13]. Thoroughly understanding the behavior of these complex phase transitions can allow us to reveal deeper principles about the Vicsek model and collective motion as a whole.

In recent years, machine learning has flourished as a field. As such, the applications of machine learning to the fundamental sciences of physics is being more thoroughly investigated [15]. Specifically, the capability of machine learning to recognize complex patterns in large datasets commonly provided by condensed matter systems has proven useful [16]. Mechanistic agent-based models are thus strong companions to machine learning for extracting patterns across a population [17], [18].

In this work, we use machine learning as a tool through unsupervised clustering for phase classifying and neural networks to investigate the phase transitions in the agent-based Vicsek model and produce a phase diagram featuring the onset of collective motion as a function of free parameters in the Vicsek model [19]. We aim to both elucidate the controversial phase boundaries in the Vicsek model and demonstrate the utility of machine learning in complex agent-based collective motion models.

The rest of this paper is organized as follows. In Sec. II, we introduce the Vicsek model, define the simulation parameters and observables used throughout this work, and summarize the relevant phases and phase transitions. In Sec. IV, we describe our methodology, including the simulation procedure, dataset construction, K-Means phase labeling, and neural-network classifier. In Sec. V, we present the resulting learned phase diagrams and analyze the predicted disorder, order, and coexistence regions. In Sec. VI, we address limitations of this study. Finally, in Sec. VII, we summarize our results and discuss future directions.

II. Background

The Vicsek model is proposed as a minimalist representation of collective motion to further simplify the mechanism of agent interactions in particle systems [9]. The model is similar to ferromagnetic models like the more recently proposed Active Ising Model [20]; however, where particles in ferromagnetic models align their spins in the same direction, particles in the Vicsek model align their orientation according to their neighbors, with the noise η becoming an analog of the temperature in ferromagnetic models and density ρ as an analog for density of spins [9].

1) Model Definition and Observables: The Vicsek model defines N self-propelled particles moving with constant speed v_0 in a two-dimensional space with linear size L, density ρ, and periodic boundary conditions. Each particle i is defined by its position $\mathbf{r}_i(t)$ and an angle $\theta_i(t)$ that specifies the direction of its velocity. Define $\mathcal{N}_i^R(t) = \{j : |\mathbf{r}_j(t) - \mathbf{r}_i(t)| < R\}$ to be the set of particles within a neighborhood of radius R of particle i at time t. Then, after each discrete time step Δt, the state $(\mathbf{r}_i, \theta_i)$ of the particle i follows the update rule

$$\begin{cases} \mathbf{r}_i(t+\Delta t) = \mathbf{r}_i(t) + v\begin{pmatrix}\cos\theta_i(t)\\ \sin\theta_i(t)\end{pmatrix} \\ \theta_i(t+\Delta t) = \mathrm{Arg}\left(\sum_{j\in\mathcal{N}_i^R(t)} e^{i\theta_j(t)}\right) + \eta\xi_i(t) \end{cases}, \quad (1)$$

where η is a noise parameter and $\xi_i(t) \sim \mathcal{U}[-1/2, 1/2]$ is a random variable chosen from a uniform distribution. The Vicsek model includes three free parameters for a fixed system size: η, ρ, and velocity v_0 [9]. Vicsek *et al.* concluded that the behavior of the model changes as η and ρ are varied, but v_0 does not affect the final-state phase in any significant way (where v_0 values from 0.003 to 0.3 were tested). However, as v_0 approaches 0, the simulation turns into an analog of the XY model [9], [21]. Additionally, varying v_0 affects the critical noise value η_c for phase transitions; as v_0 approaches 0, particles stay in their local neighborhood longer, which allows for better local alignment but overall worse spread of alignment information across the system and decreases η_c, which suggests that systems with lower v_0 are less able to sustain order [10].

In this work, we use several key metrics detailed below. The order parameter ϕ, or the normalized magnitude of the sum of all particle velocities, is calculated as

$$\phi(t) = \frac{1}{Nv_0}\left|\sum_{i=0}^{N} v_i(t)\right|, \quad (2)$$

is recorded every 10,000 time steps for usage in the dataset construction, and measures the degree of global alignment at any given point in the simulation run. The order parameter will vary in [0, 1], with 0 representing a state of perfect order and 1 representing a state of perfect disorder [9].

With the order parameter, we calculate the stationary order parameter $\langle\phi\rangle$, the long-term order parameter once the simulation has reached a steady state, with

$$\langle\phi\rangle = \lim_{T\to\infty}\frac{1}{T}\int_0^T \phi(t)\,\mathrm{d}t\,, \quad (3)$$

where $\langle\phi\rangle$ represents the time-averaged order parameter [1]. In this formula, we represent T as the number of time steps the order parameter is averaged over, thus proposing $\langle\phi\rangle$ as a final-state order parameter. When calculating $\langle\phi\rangle$, we discard the transient phase of the simulation, or the first 20% of the simulation, thus counting only steps 200,000 onwards. The stationary order parameter thus maps the final state of the simulation as a function of η, ρ, and v_0, with the same range as the order parameter ϕ.

We also calculate the Binder Cumulant G, a measure of the fluctuation of the order parameter, with

$$G = 1 - \frac{\langle\phi^4\rangle}{3\langle\phi^2\rangle^2}. \quad (4)$$

The Binder Cumulant, as a measure of the fourth moment of the order parameter ϕ, is commonly used in finite-size scaling to find the critical noise value η_c with varying system sizes L [2], [22]. In this work, we use the Binder Cumulant to identify phase separations in our constructed dataset.

In addition, we calculate the susceptibility χ, or variance in the order parameter, as

$$\chi = N\langle(\phi^2 - \langle\phi\rangle^2)\rangle. \quad (5)$$

We utilize susceptibility for its exhibited behavior at phase transitions, where the χ value peaks at continuous phase transitions.

We use these metrics when constructing the dataset to further analyze the phases and phase transitions in the Vicsek model.

2) Phases and Phase Transitions: The Vicsek model exhibits a transition in its final state as the free parameters of the model are varied. It can be seen that decreasing the free parameter of noise η while keeping the other free parameters of ρ and v_0 constant below a critical value η_c leads to a change in the final state of the simulation from disorder to polar order, where the particles have aligned their directions [1]. Similarly, changing the free parameter ρ while keeping other free parameters constant across a critical value will lead to the final state of the ordered phase at high values of ρ and disordered phase at low values of ρ.

In his original paper, Vicsek *et al.* [9] concluded that there is a second-order continuous transition from the disorder phase, which was characterized by low noise or high density, to the polar ordered phase, which was characterized by high noise or low density. In the ordered phase, particles align their direction of motion with $\phi \approx 1$ and $G \approx 2/3$. The ordered phase is similar to the liquid phase in the liquid-gas model as well, which has contributed a number of discoveries regarding phases and their transitions in the Vicsek model [12]. The existence of the long-range order phase with number of dimensions $d = 2$ was also verified by Toner and Tu with a set of hydrodynamic equations [6].

The disordered phase, the original counterpart to the ordered phase, appears as a system of particles moving in random, uncorrelated directions, and is characterized by high noise or low density. As particles are unable to align with their neighbors either due to high noise or low density, the disordered phase appears like a gas-like phase [23], with $\phi \approx 0$ and $G \approx 0$.

However, the nature of the phase transition was later disproved a decade later using finite-size scaling with the Binder Cumulant to be first-order, with the original conclusion from Vicsek *et al.* attributed to strong finite-size effects in his experiments [2]. The critical values of noise η_c and ρ_c were also shown to be affected by varying v_0 [24].

Later, the existence of a third phase of coexistence was discovered by Solon *et al.* through the interpretation of the Vicsek model as a liquid-gas model, with the behaviors of metastability and nucleation observed close ot the phase boundaries [12]. The coexistence phase, or banding phase, is characterized by a set of ordered traveling periodic bands traveling at a speed the order of v_0, which can be considered as a polar liquid amongst a disordered gas background of particles between the traveling bands [13], [25]. In the coexistence phase, both ϕ and G tend to middling values between those of the ordered and disordered phases. Most notably, the susceptibility χ peaks in the coexistence phase around phase transitions, which aids to the distinction between phases in the K-Means clustering. In the coexistence phase, the bands are randomly spaced in the standard Vicsek model. However, when the final-state is allowed to relax for an extended transient, possibly many millions of steps, the phase exhibits regularly spaced traveling bands [12]. When the thermodynamic limit is approached, as $N \to \infty$ and $L^2 \to \infty$ to render fluctuations negligible and apply hydrodynamic theories, the traveling bands exhibit properties that are independent of the density ρ and system size L. When varying ρ, the number of bands is the only change in the phase, with the size and speed of the traveling bands remaining the same [12]. However, as the order-coexistence boundary is approached, the number of traveling bands increases significantly, producing a chaotic, highly crowded state, or smectic arrangement, near the phase boundary. As such, the distinct pattern here invites the possibility of a split of the coexistence phase into non-smectic and smectic phases [25].

The fourth cross-sea phase was introduced by Kürsten *et al.* for extremely large system sizes and select parameter ranges at the edge of the ordered-coexistence phase transition [13]. The cross-sea phase, named for the ocean phenomenon of grid-like waves particularly treacherous to sailors, exhibits a new non-smectic complex pattern independent of that seen in the coexistence phase that can also be observed in the natural cross sea pattern in oceans [13], [19].

In this work, we will focus on analyzing the ordered, coexistence, and disorder phases to produce a coarse phase map of the Vicsek model.

III. Application

Phase transitions are a fascinating part of our natural world that are crucial to understand for our comprehension of complex systems and their inherent principles (cite the pharmeceutical journal). As one of the most simple collective motion models that exhibits a liquid-gas phase transition [11], the Vicsek model is an excellent subject for study to further understand the universal rules of collective motion in our universe [1], [2]. With our understanding from the inception of collective motion in the Vicsek model, we can apply this to universal systems and purposes. For example, specific application areas that use phase-transition-induced techniques are modern technologies called phase-change materials [26]. These are materials that undergo phase transitions to take advantage of different phase properties for objectives like better smart textiles and energy storage systems.

IV. Methods

In order to effectively detect phase transitions in the Vicsek model, we aim to establish a clean global picture of where the different collective states exist. Using an optimized version of the Vicsek model, we build a labeled dataset with varying η, ρ, and v_0 between instances, with each instance labeled with the dominant phase at the final state of the simulation. We then apply a neural network classifier to this dataset to produce a global phase map with phase transitions.

A. Simulation

In this study, we use a basic numbers-only C++ implementation of the Vicsek model described above to optimize computational efficiency. We keep all parameters constant across runs barring the free variables η, ρ, and v_0 with a consistent system size of $L = 128$. Each simulation is run to 10^6 time steps to ensure final-state convergence.

In our coded implementation of the Vicsek model, the three free variables are passed at runtime, which allows for variation in parameter values in order to construct the dataset more efficiently. The other parameters are fixed with $L = 128$, $r_0 = 1.0$, and $\Delta t = 1.0$. Each particle is placed randomly at the beginning of the simulation over simulation space $(0, L)^2$, and orientations selected randomly between $[-\pi, \pi)$. This produces a state of complete disorder at the start of the simulation, which then converges to its final state based on the simulation's free parameters.

Using a method developed for the study of molecular dynamics, we utilize a cell list to minimize $O(n^2)$ neighbor calculations [1]. We partition the simulation space into cells of size r_0. Each particle is hashed into a cell, and we utilize a linked list to efficiently retrieve all particles in a cell, which reduces neighbor-finding complexity from $O(n^2)$ to $O(n)$.

Additionally, to minimize computing time with intensive simulation runs, we utilize High-Performance Computing (HPC). In our code, we use OpenMP [27], a parallel computing API, for three separate computationally expensive loops: the orientation update, position update, and order computation loops. In this way, computations are distributed across multiple threads, which allows for the parallel execution of code on different CPU cores. The use of OpenMP allows us to parallelize tasks across all available CPU cores, which significantly lowers computational expenses when used in conjunction with the TJHSST Supercomputer Cluster, where each simulation run is processed through the TJHSST Supercomputer Cluster across 5 nodes with 8 CPUs per node [27], [28]. In this way, we are able to run simulations and computer order parameters for each simulation in a more efficient and streamlined fashion.

B. Dataset Construction

In our dataset, we aim to effectively represent points in the 3-dimensional (η, ρ, v_0) space with a special focus on regions with phase transitions. To achieve this, we use Latin Hypercube Sampling [29] to sample parameters in a way that captures the essential structure of the space without wasting valuable computation time in unimportant areas. With Latin Hypercube Sampling, we split each dimension into k ordered intervals for k samples, and select a random point from each interval. This ensures that only one point is selected per hypercube, thus guaranteeing an even distribution overall for any combination of variables. We choose our parameter ranges as $\eta \in [0, 1]$, $\rho \in [0.1, 3]$, and $v_0 \in [0.01, 0.5]$, with ρ and v_0 on a $\log_{10}$ scale to cover the ranges evenly, as they span multiple orders of magnitude, rather than allowing large values to dominate our sampled values.

For each chosen parameter point $\lambda = (\eta, \rho, v_0)$, we run 5 simulation trials with random initial conditions. In order to assign phase labels to each parameter point in our dataset, we turn to unsupervised machine learning clustering algorithms. We calculate the stationary order parameter $\langle\phi\rangle$, Binder Cumulant G, and susceptibility χ per each simulation-run parameter point λ trial, which allows us to produce 5 separate trial datasets in $(\langle\phi\rangle, G, \chi)$ space, with consistent indexing across datasets aligning with the (η, ρ, v_0) representation of each parameter point.

We find that the K-Means algorithm [30] with 3 centroids produces the most well-formed clusters out of DBSCAN [31], GMM [32], Spectral Clustering [33], and K-Means with varying centroids in (1, 5). The efficacy of K-Means is shown through the three intrinsic performance metrics of the Silhouette Score [34], Davies-Bouldin Index [35], and Calinski-Harabasz Index [36]. Each clustering algorithm was tested on a dataset formed from averaged $(\langle\phi\rangle, G, \chi)$ values across the 5 trials for each parameter point and evaluated based on the intrinsic performance metrics. We found that the K-Means algorithm with 3 centroids performed the best out of the tested clustering algorithms with the highest Silhouette Score, lowest Davies-Bouldin Index, and highest Calinski-Harabasz Index.

Before applying the K-Means algorithm to our data, we first normalize the dataset features for each dataset using z-score normalization to ensure all features contribute equally to K-Means distance calculations with the formula

$$Z = \frac{X - \mu}{\sigma}, \tag{6}$$

where Z represents the standardized data point, X represents the raw data point, μ represents the average of that feature across the dataset, and σ represents the standard deviation of that feature across the dataset.

Once the features are normalized, we apply the standard K-Means algorithm to the trial datasets with 3 centroids selected with K-Means++ initialization [37], where centroids are selected according to a probability formula

$$p(x) = \frac{d^2(x)}{\sum_{x' \in \mathcal{N}} d^2(x')}, \tag{7}$$

where d represents the Euclidean distance from a data point x to the nearest centroid and $\mathcal{N}$ represents the set of all points.

In Fig. IV-A, we can see the K-Means clustering results in $(\langle\phi\rangle, G, \chi)$ space, where the clusters were formed, and their representation in (η, ρ, v_0) space in Fig. IV-A. We can see how representation of parameter points in $(\langle\phi\rangle, G, \chi)$ space leads to distinct, separated groups that are classified effectively by the K-Means algorithm. The clustered disordered phase, marked in purple, is characterized by an extremely low order parameter ϕ, a relatively low Binder Cumulant G, and extremely low susceptibility χ. However, the coexistence phase, marked in blue, composes a lower proportion of the overall data and is characterized especially with a high susceptibility χ near phase transition points, a middling order parameter ϕ, and a relatively high Binder Cumulant G. Lastly, the ordered phase marked in orange is characterized by a relatively high order parameter ϕ, Binder Cumulant extremely close to 0.66, and an extremely low susceptibility. These parameter trends align with our expectations denoted in **??**, which suggests that our proposed K-Means framework does effectively capture the phase structure and underlying dynamics in the Vicsek model.

Using the nominal predicted K-Means cluster labels, where 0 refers to the disorder phase, 1 refers to the order phase, and 2 refers to the coexistence phase, we label each parameter point in $\lambda = (\eta, \rho, v_0)$ representation with the dominant phase present in the simulation's final state with

$$\mathrm{phase}(\lambda) = \mathrm{argmax}_k \, n_k(\lambda), \tag{8}$$

where n_k is the number of the 5 trials that ended in phase k. With these methods, we form a dataset with 3 independent features, 193 instances, and 3 main nominal labels (representing the order, coexistence, and disorder phases).

C. Neural Network Classifier

After assigning phase labels to the sampled parameter points, we train a neural network classifier to learn the map

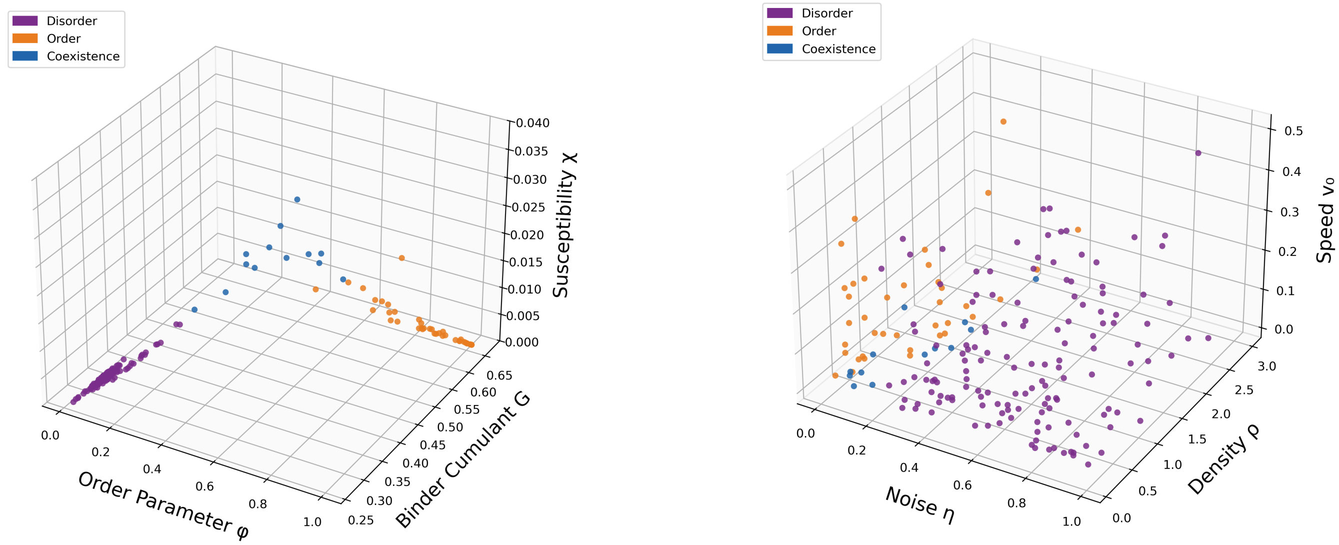

Fig. 1. K-Means classification of Vicsek-model simulation data. (a) Clustering in the observable space spanned by the order parameter $\langle \phi \rangle$, Binder cumulant G, and susceptibility χ. (b) The same cluster labels projected onto the parameter space (η, ρ, v_0), showing the corresponding disorder, order, and coexistence regions.

from Vicsek-model parameters to long-time phase behavior. Each input instance is represented by the three-dimensional feature vector $\mathbf{x} = (\eta, \rho, v_0)$, and the corresponding target label is the dominant phase assigned during the clustering procedure described above.

The labeled dataset is split into training and testing sets, with 80% of the data used for training and 20% reserved for testing. Because the phase labels are not evenly distributed across the dataset, we use a stratified train-test split so that the relative proportion of each phase is approximately preserved in both sets. This prevents underrepresented phases from being disproportionately excluded from either the training or testing data.

We use a fully connected feed-forward neural network with one hidden layer. The network takes the three simulation parameters (η, ρ, v_0) as input, passes them through a hidden layer with 128 neurons and a ReLU activation function, and then outputs three scores corresponding to the three possible phase labels. To reduce overfitting, dropout with probability 0.2 is applied after the hidden layer. The classifier therefore has the form

$$f(\eta, \rho, v_0) = \mathbf{z} = (z_0, z_1, z_2) \tag{9}$$

where z_k is the network score associated with phase k. These three scores are converted into probabilities using the softmax function,

$$P(k|\eta, \rho, v_0) = \frac{e^{z_k}}{\sum_j e^{z_j}}, \tag{10}$$

where the sum runs over the three possible phase labels. The predicted phase is chosen as the phase with the largest probability:

$$\hat{k} = \mathrm{argmax}_k \, P(k|\eta, \rho, v_0). \tag{11}$$

The model is trained using a weighted cross-entropy loss. Since the dataset is imbalanced, the loss for each class is weighted according to the inverse frequency of that class in the training set. Specifically, if N_{train} is the number of training examples, $K = 3$ is the number of classes, and N_k is the number of training examples in class k, then the weight assigned to class k is

$$w_k = \frac{N_{\mathrm{train}}}{K N_k}. \tag{12}$$

This weighting penalizes mistakes on underrepresented classes more strongly, preventing the classifier from simply favoring the most common phase. Training is performed with the Adam optimizer using a learning rate of 10^{-3}, a batch size of 32, and 200 training epochs. During training, we monitor both the training accuracy and the test accuracy, which provides a measure of how well the classifier generalizes to parameter points that were not used during training.

Once trained, the classifier is applied to a regularly spaced grid of points in the (η, ρ, v_0) parameter space. For each grid point, the network predicts one of the three phase labels using the same largest-probability rule in Eq. (11). The resulting labeled grid is then used to construct phase-map visualizations. In this way, the neural network interpolates between the simulated data points and produces an approximate global phase diagram for the Vicsek model.

V. Results

We produce a global coarse phase map in (η, ρ, v_0) space with the help of the neural network and labeled dataset. In Fig. 2, we see the three-dimensional grid classified by the neural network that produces our phase diagram of the Vicsek

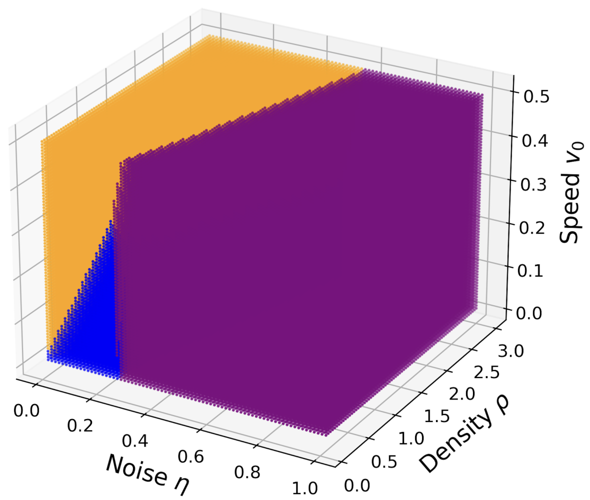

Fig. 2. Three-dimensional learned phase diagram of the Vicsek model in the parameter space (η, ρ, v_0). Each point represents a classified parameter-grid point, with purple, orange, and blue denoting disorder, order, and coexistence, respectively. The classifier predicts a broad ordered region at low noise, a disordered region at higher noise, and a narrower coexistence region separating the two phases.

model. The neural network classifier assigns a nominal label to each instance with a testing classification accuracy of 0.92. In Fig. 2, we represent the classified labels with colors: orange for order, blue for coexistence, and purple for disorder. We can note the existence of the narrow coexistence band around $\eta = 0.0$ to $\eta = 0.2$, surrounded by the ordered phase and disordered phase. From this visual, we can denote the effect of the free variables η, ρ, and v_0 on the final dominant phase of each simulation, as seen from the phase boundary lines. While η appears to be the greatest contributing parameter to phase transitions, we can note that varying v_0 separates the ordered and coexistence phases, with ρ affecting the boundary between the ordered and disordered phase at high values of v_0.

Additionally, we can see the phase boundaries in greater detail with Fig. 3, across the two-dimensional (η, ρ) slices at varying v_0 values. As η dips below a critical noise value $\eta_c = \eta_c(\rho, v_0)$ determined by the values of the other free parameters ρ and v_0, we observe a phase transition from the disordered phase to the coexistence phase to the ordered phase. We can also observe that higher values of ρ increases the critical noise threshold η_c, thus showing that systems with higher ρ values are able to sustain order with higher η values. These observations are consistent with the known behavior of an increasing noise threshold η_c with ρ. Additionally, higher values of v_0 reduce the width of the coexistence phase band, or the range of η values classified as coexistence, due to an increased number of interactions between particles and decreased number of noise fluctuations that result in the coexistence phase.

VI. Limitations

The greatest limitation on this study came from computational power. As we were only able to run 5 jobs on the TJHSST Supercomputer Cluster [28], the dataset took a long to construct since each run took many hours, and we were running thousands of runs. This is why we were not able to produce the cross-sea phase [13] and also why the dataset was limited to 193 instances.

VII. Conclusion

In this work, we used machine learning as a tool to construct a phase diagram of the Vicsek model in (η, ρ, v_0) space featuring the by utilizing the K-Means clustering algorithm and an artificial neural network. Using an HPC-optimized version of the Vicsek model, we built a 193-instance dataset of simulation parameter points in (η, ρ, v_0) with Latin Hypercube Sampling by running 5 trials per parameter point and recording the metrics of the stationary order parameter $\langle \phi \rangle$, Binder Cumulant G, and susceptibility χ from each trial. We then utilized the K-Means clustering algorithm to assign labels to the dataset by clustering the parameter points using their representation in $(\langle \phi \rangle, G, \chi)$ space. With the clustered labels across the 5 trials, we assign a dominant phase to each instance in our dataset. By applying a feed-forward neural network to our constructed dataset, we are able to classify a regularly spaced grid of points in (η, ρ, v_0) space, which is then used to construct our phase map. We can conclude that machine learning is an effective tool for analysis in complex agent-based collective motion models through usage of both unsupervised clustering algorithms to classify phases and a neural network to classify and investigate the phase transitions.

VIII. Future Work and Recommendations

The research we detail in this paper has a number of possible future extensions. The coarse phase map produced in this work sets the context for phase boundaries in parameter space, which allows us to further analyze phase transitions in the Vicsek model by studying specific regions where the dominant label predicted by the neural network in this paper changes between neighbors in parameter space. Instead of using a classifier to predict separate nominal phases in free parameter space, we can map the probability of each phase at a certain parameter point, where we consider separate phases as nonequilibrium attractors. Then, our hard phase classification turns into a classification based on basin-occupation probability of each attractor. We can utilize the phase boundaries found in this work to sample parameter points more densely around the boundaries to further study coexistence of multiple phase attractors. Then, we can replicate the methods in this work, where we produce a labeled dataset consisting of parameter points in regions where the system is highly sensitive to initial pertubations, and use this dataset to train a neural network to visualize multistability in different regions, extract a probabalistic boundary, and identify areas of coexistence regions. This shift of perspective from studying the separate phase to analyzing the coexisting attractors at phase boundaries provides a novel and useful framework for the larger community, and a baseline for a further extension of this paper.

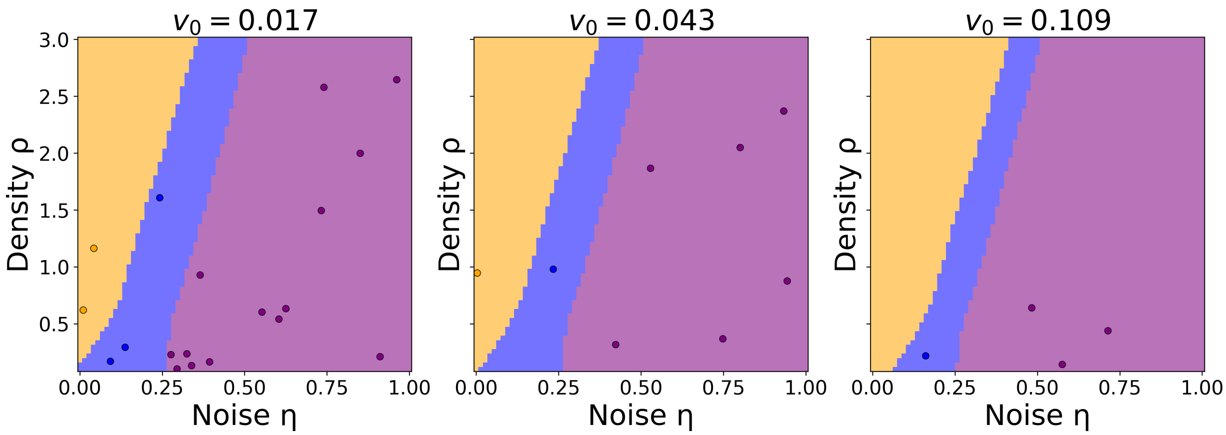

Fig. 3. Two-dimensional slices of the learned Vicsek-model phase diagram in the (η, ρ) plane at fixed particle speeds $v_0 = 0.017$, 0.043, and 0.109. The shaded regions show the neural-network classification on an evenly spaced parameter grid, while the overlaid points show the original labeled simulation data near each slice. Purple, orange, and blue denote disorder, order, and coexistence, respectively.

An additional possible extension of this work focuses on properties of hysteresis near the Vicsek model's phase transitions. In order to observe the exhibition of hysteresis in the model, we can observe how the basin-occupation probabilities of coexisting attractors change based on history. Hysteresis, a phenomenon that causes the final state of the Vicsek model to depend on the initial random scattering of agents given the same simulation parameters, can be observed when downwards parameter sweeps and upwards parameter sweeps towards the same parameter values differ based on history. Using η values from the edge of coexistence bands in the previous extension work, we can train another NN on a constructed dataset of steady-state simulations to produce probability landscapes, and identify hysteresis as a geometric difference betwen the upwards-sweep and downwards-sweep landscapes.

Overall, our research produces results that provide more insight into phase transitions in Vicsek model as a whole and validate the efficacy of the applications of machine learning to mechanistic agent-based models. These results also introduce more applications of machine learning to the Vicsek model in particular, with a special focus on phase transition regions provided by the coarse phase map produced in this work.

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